

Harshana Lakshara Fernando

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SUMMARY

Software Engineer with production experience building Java/Spring Boot microservices, Kafka-backed distributed systems, and gRPC integrations. Delivered backend services and rollout strategies for production systems, including a Kafka-backed PNR generation service handling 25+ PNRs/sec at <100ms latency, with additional experience in PostgreSQL, Go/Gin, Docker, Kubernetes, CI/CD, and observability.

EXPERIENCE

R&D Software Engineer — KōreroCare Platform Mar 2026 – Present
Empathic Computing Laboratory, University of Auckland Remote

- Co-led backend design for the **KōreroCare** mobile-wearable sensing platform, building **Go/Gin**-based services for time-series ingestion and cross-device synchronization.
- Designed **PostgreSQL** schemas for participants, studies, devices, consent, reminders, and wearable sensor streams.
- Integrated **Keycloak OAuth2/OIDC** with **Traefik** gateway routing and Docker Compose deployment for secure internal service access.
- Developed Firebase notification and reminder flows for participant engagement and study prompts.
- Modeled an **HL7/FHIR**-aligned data schema mapping participants, devices, and wearable measurements.

Software Engineer Intern — ISA Aviation Technology Dec 2024 – May 2025
Information Systems Associates (Pvt) Ltd Colombo, Sri Lanka

- Owned design and delivery of a distributed **Spring Boot/Kafka**-backed PNR generation service, **live in production**, achieving **25+ PNRs/sec** at **<100ms latency**.
- Led the rollout from legacy PNR generation to AeroOrder using database-driven **feature toggles** and a **strangler-pattern** migration approach.
- Resolved production **Kafka consumer lag** and offset-reset issues by adding graceful shutdown hooks, retry handling, validation, and a Producer-Consumer refactor.
- Integrated **gRPC** clients and Kafka consumers into a legacy **Java 8** AeroMart monolith for backward-compatible communication with AeroOrder services.
- Built load-testing and monitoring support using **JMeter**, **Prometheus**, and **Grafana** to validate async order/PNR flows, queue time, and service latency.
- Added Jenkins environment support and maintained **>90% unit test coverage** while coordinating delivery with QA and distributed engineering teams.

Freelance Software Engineer — Selected Client Projects May 2025 – May 2026
Client Platforms Remote / Sri Lanka

- Delivered **AgriNova**, a multi-branch inventory and sales platform, owning backend workflows, deployment, and client-facing fixes end-to-end.
- Implemented transaction-safe stock logic across **3 core workflows**: material issuance, returns, and purchase operations.
- Resolved customer-reported production issues including authentication/version compatibility problems and workflow-level validation bugs.
- Reduced repeat dashboard data retrieval latency from **~384ms** to **<1ms** by implementing Next.js/React Query caching, pagination, and **PostgreSQL** indexing.
- Set up deployment and CI/CD workflows for full-stack delivery using Vercel and Supabase/PostgreSQL.

PROJECTS

Employee Management System [Architecture] Java 17, Spring Boot, Spring Cloud, Microservices, Kubernetes
Built cloud-native microservices using Eureka, Gateway, RabbitMQ-based config refresh, and Resilience4j (**Circuit Breaker**, **Rate Limiter**).

Serverless Auction Platform [Architecture] Node.js, AWS Lambda, API Gateway, DynamoDB, SQS/SNS
Architected event-driven auction/auth services using **AWS Lambda**, EventBridge, **DynamoDB** GSIs, and custom JWT authorizers.

ZK-Rollup L2 Benchmarking Framework Rust, Solidity, TypeScript, Python, Docker
Built benchmark pipelines with **Rust** Submitters and **Solidity** contracts, measuring **~70% lower DA cost** while raising coverage from **41% to 88%**.

More projects, architecture diagrams, and articles are available on my portfolio.

SKILLS

Backend: Java, Spring Boot, Spring Cloud, Microservices, Go/Gin, Node.js, REST, gRPC, GraphQL, Hibernate

Messaging & APIs: Kafka, RabbitMQ, WebSockets, Protobuf, JWT, OAuth2/OIDC

Data: PostgreSQL, MySQL, Oracle DB, Redis, Couchbase, DynamoDB, Prisma

Cloud / DevOps: AWS, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD

Observability / Testing: Prometheus, Grafana, JMeter, Gatling, JUnit, Mockito, Pytest

EDUCATION

B.Sc. (Hons) in Engineering — Computer Science and Engineering, University of Moratuwa 2022 – 2026
G.C.E. A/L: 3 A's, Physical Science Stream, District Rank 17

TRAINING & CERTIFICATIONS

CKA Training (KodeKloud, 2025) · AWS SAA-C03 Training (Udemy, 2025) · Apache Kafka Series (I, II, III) · 100 Days of DevOps (KodeKloud, 2025) · Advanced OAuth Security (Udemy, 2024)